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1 SMART PENDANT To Monitor Child Immunity

India is valiantly fighting the curse of high numbers of neonatal deaths and maternal mortality rate (MMR). A study done by UNICEF in 2018 revealed that almost 25 per cent of the global neonatal deaths occur in India. While this number is reducing, it is still quite alarming. Moreover, children of all ages in villages and semi-urban areas are always at risk, as nearly 1.5 million children fall victim to diseases every year. All these cases can be prevented with proper immunisation and vaccination drives.

MMR—death of the mother during or post-parturition—is also something to be taken very seriously. Khushi Baby, a non-profit organisation (NGO) with teams based out of the US and India, have combined near field communication (NFC), cloud and data analytics to create a digital necklace that will help healthcare workers in semi-urban and rural areas to keep mothers and children safe.

Traditionally, community health workers visit homes to collect records about necessary immunisations done to infants. These records are stored manually in physical registers, which often become inconvenient for reference and are prone to damage. The complete Khushi Baby system includes a durable and waterproof necklace with an NFC chip inside a pendant, which interfaces with the proprietary mobile app developed for community healthcare workers.

In place of sensors or power supply, the chip in the pendant incorporates an encrypted data stor-

age device (up to 888 bytes), which is readable and writable, and stores essential data from the vaccination card.

Infants are given the necklace to wear, while healthcare workers record the necessary immunisation data over

the mobile app. Data is recorded using a phone with inbuilt NFC reader/writer, which can read and update information stored on the chip. When healthcare workers are in the presence of a network, they can upload the collected data to the cloud. This

